

BATCHING ALARMS

NAME	DESCRIPTION	NOTES
Batching time-out	It activates after 4 seconds from the start command, when any flow is detected	When alarms activate the actuator close automatically
Overflow	It activates when the batching quantity is more than 20% of the set one.	Both alarms can be visualized remotely, setting the usc2 function on "HARD W. AL" or "ALL ALARMS"

Visualize alarms via display:

The presence of a dosage alarm is highlighted by the message: "ALARM"

Visualize alarms via MCP:

Batching page: The presence of a batching alarm is highlighted by the message: "ALARM"

Alarms page: The presence of a batching alarm is highlighted by the message: "BATCHING ERR."

ATTENTION!!**DISABLED FUNCTIONS WHEN BATCHING IS ACTIVATED**

["1.19" pag27](#)

["4.3" pag28](#)

["4.4" pag28](#)

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(POS. 7.3-4) Current output option and range [Out mA1/2=X_XX XXX] AL3 [AO1CF] [AO2CF]

This function sets the current output 1 and 2. This function is optional and will not appear unless the option has been requested. There are three fields to modify for this function:

☐ Scale zero: 4 or 0mA

☐ Full scale: 20 or 22mA

☐ Field: + = positive, - = negative, blank = both, -0+ = central zero scale

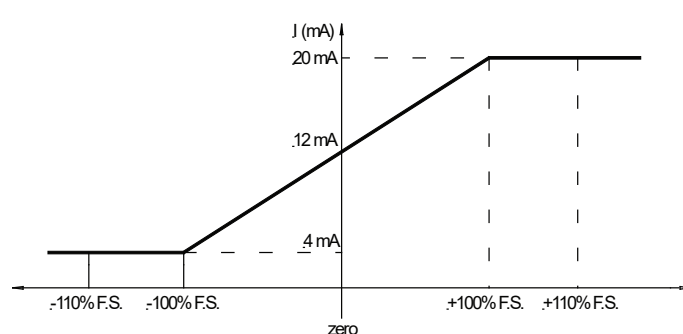
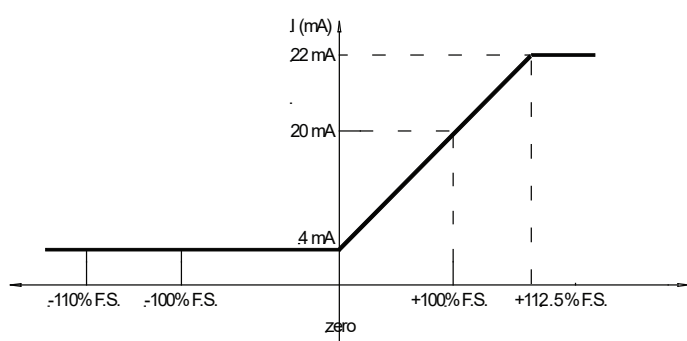
The values corresponding to the scale points are shown in the following chart:

CURRENT VALUES IN mA ASSOCIATE TO THE % FULL SCALE VALUE					
POSSIBLE FIELD	REVERSE FLOW VALUE		ZERO	DIRECT FLOW VALUE	
	≤-210%	-100%		+100%	≥+210%
Out.mA = 0 ÷ 20 +	0	0	0	20	20
Out.mA = 0 ÷ 22 +	0	0	0	20	22
Out.mA = 4 ÷ 20 +	4	4	4	20	20
*Out.mA = 4 ÷ 22 +	4	4	4	20	21.6
Out.mA = 0 ÷ 20 -	20	20	0	0	0
Out.mA = 0 ÷ 22 -	22	20	0	0	0
Out.mA = 4 ÷ 20 -	20	20	4	4	4
Out.mA = 4 ÷ 22 -	21.6	20	4	4	4
Out.mA = 0 ÷ 20	20	20	0	20	20
Out.mA = 0 ÷ 22	22	20	0	20	22
Out.mA = 4 ÷ 20	20	20	4	20	20
Out.mA = 4 ÷ 22	21.6	20	4	20	21.6
Out.mA = 0 ÷ 20 -0+	0	0	10	20	20
Out.mA = 0 ÷ 22 -0+	0	1	11	21	22
**Out.mA = 4 ÷ 20 -0+	4	4	12	20	20
Out.mA = 4 ÷ 22 -0+	2	4	12	20	22

In hardware alarm conditions "HARDW AL." (interrupted coils, empty pipe, measure error) the current value is programmed by the function "mA v.fault" (pos. 5.6) and it is expressed as percentage of a fixed current range, where: 0% = 0mA and 210% = 22mA.

* Example 1: out 4-22 +

** Example 2: out 4-20 -0+



(POS. 7.5) Analog Output1 full scale

[A1S= dm/s x.xxxx]

AL3 [AO1FS]

It allows to set the full scale value for analog output 1 **independently** from the main scale of the instrument.

(POS. 7.6) Analog Output2 full scale

[A2S= dm/s x.xxxx]

AL3 [AO2FS]

It allows to set the full scale value for analog output 2 **independently** from the main scale of the instrument.

MENU 8 - COMMUNICATION

(POS. 8.1) Hart Preamble	[HART pr.=XXXXXX]	AL3	[HARTP]
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Number of preamble (Hart function)

(POS. 8.2) Hart O.C	[HART O.C=ON/OFF]	AL3	[HRTOC]
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Hart bus output control 4-20mA.



(POS. 8.3) Device Address	[Dev. Addr=XXXXXX]	AL3	[DVADD]
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Device communication address number.

(POS. 8.4) Modbus Speed	[Speed=XXXXXX]	AL3	[MDBSP]
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Modbus link speed.

(POS. 8.5) Modbus Parity	[Parity=XXXXXX]	AL3	[MDBPA]
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Modbus link parity.

(POS. 8.6) Modbus Delay	[Delay=XXXXXX]	AL3	[MDBDL]
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Modbus replay delay.

(POS. 8.7) Modbus chars Timeout	[C. timeout=X]	AL3	[MDBCT]
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Maximun delay between chars (frames).

MV210 MODBUS PROTOCOL

☐ RS485 HARDWARE CONNECTION

For the hardware connection see the relative section in this manual and MODBUS manual.

☐ DATA WORD FORMAT

The data bytes travelling in serial form on the communication line are enclosed in words which have a fixed length of 10 bits:

1 START BIT

8 DATA BITS = 1 DATA BYTE

1 STOP BIT

Each word contains one byte of data plus additional bits which serve to synchronies and make the communication safer. These extra bits added automatically in the transmission phase by the transmitter integrated circuit. In the reception phase, the reverse operation is executed by the receiver integrated circuit: the eight data bits are extracted and the others eliminated. These operation are executed entirely on a hardware level. The 8 data bits must be serialised staring from bit 0 (the least significant one).

☐ COMMUNICATION SPEED

The millennium series instruments have 4 communication speeds:

4800 bps

9600 bps

19200 bps

38400 bps

□ SERIAL PORT SETTINGS

Serial port setting:

Data bits: 8

Parity: Manu < 7-Communication >, function - < Parity >

Stop bits: 1

Flow control: none (no control lines no xon/xoff characters used)

□ GENERAL DESCRIPTION

All data are sent in group of 16 bits registers. The format used is BIG ENDIAN, MSB byte is sent first, LSB is sent last. When a variable is more than 16 bits in size, it uses two adjacent registers. The totalizer values are expressed as integer numbers. For the correct representation of the value in case there is a fractional part, the decimal dot must be placed in the position specified by the next variable register following the totalized value. All values relative to the flow rate are averaged. The number of samples that compose the average value varies depending on the measure sample rate and the MODBUS reading requests. Example: measure sample rate = 50 Hz, MODBUS reading frequency = 10 Hz, Number of samples used for average calculation = $50/10 = 5$.

Read process variable : COMMAND 04.

ADDRESS	SIZE	TYPE OF DATA	MEANING
0000-0001	2 REGISTERS 32 BITS	FLOAT	full scale flow rate in the unit of measure chosen (as can be seen in the display of the instrument)
0002-0003	2 REGISTERS 32 BIT	FLOAT	flow rate value in percentage
0004-0005	2 REGISTERS 32 BITS	FLOAT	flow rate value in the unit of measure chosen
0006-0007	2 REGISTERS 32 BITS	FLOAT	Flow speed in the unit of measure chosen (m/s or ft/s)
0008-0009	2 REGISTERS 32 BITS	UNSIGNED LONG	Totalizer T+ value
0010	1 REGISTER 8+8 BITS	TWO BYTES	First byte (MSB): number of overflows recorded, second byte (LSB): number of decimal places
0011-0012	2 REGISTERS 32 BITS	UNSIGNED LONG	Totalizer P+ value
0013	1 REGISTER 8+8 BITS	TWO BYTES	First byte (MSB): number of overflows recorded, second byte (LSB): number of decimal places
0014-0015	2 REGISTERS 32 BITS	UNSIGNED LONG	totalizer T- value
0016	1 REGISTER 8+8 BITS	TWO BYTES	First byte (MSB): number of overflows recorded, second byte (LSB): number of decimal places
0017-0018	2 REGISTERS 32 BITS	UNSIGNED LONG	Totalizer P- value
0019	1 REGISTER 8+8 BITS	TWO BYTES	First byte (MSB): number of overflows recorded, second byte (LSB): number of decimal places
0020	1 REGISTER 8+8 BITS	TWO BYTES	first byte (MSB): process flags 1, (LSB): process flags 2
0021	1 REGISTER 16 BITS	UNSIGNED SHORT	Number of measure samples used for to calculate the latest read average value of flow rate
0022	1 REGISTER 16 BITS	UNSIGNED SHORT	Equivalent resistance measured between electrode E1 and the common point, in kilo ohm
0023	1 REGISTER, 16 BITS	UNSIGNED SHORT	equivalent resistance measured between electrode E2 and the common point, in kilo ohm
0024	1 REGISTER, 16 BITS	SIGNED SHORT	voltage measured between electrode E1 and the common point, in millVolts
0025	1 REGISTER, 16 BITS	SIGNED SHORT	voltage measured between electrode E2 and the common point, in millVolts
0026	1 REGISTER, 16 BITS	UNSIGNED SHORT	voltage measured at rechargeable terminals, in milliVolts
0027	1 REGISTER, 16 BITS	UNSIGNED SHORT	residual battery capacity in percentage
0028	1 REGISTER, 16 BITS	SIGNED SHORT	CPU temperature in the unit of measure chosen
0029	1 REGISTER, 16 BITS	SIGNED SHORT	Board temperature T1 the unit of measure chosen
0030	1 REGISTER, 16 BITS	SIGNED SHORT	Board temperature T2 the unit of measure chosen
0031	1 REGISTER, 16 BITS	SIGNED SHORT	Flow sensor coil's temperature the unit of measure chosen
0032	1 REGISTER, 16 BITS	UNSIGNED SHORT	latest sensor test result code
0033	1 REGISTER, 16 BITS	UNSIGNED SHORT	number of alarms currently active

Meaning and value of the process flags 1 returned with the register 0020 (MSB):

- bit 7 (MSB): flow rate alarm MIN (flow rate below the minimum threshold set)
- bit 6: flow rate alarm MAX (flow rate over the maximum threshold set)
- bit 5: flow rate sign (1 = negative)
- bit 4: flow rate below the cut-off value
- bit 3: measure range active (0= range 1, 1= range 2)
- bit 2: flow rate measure reset value status (1= measure is forcibly reset to zero)
- bit 1: volume counters lock status (1= counters are locked)
- bit 0 (LSB) : internal use, no meaning

Meaning and value of the process flags 2 returned with the register 0020 (LSB):

- ☐ bit 7 (MSB): flow rate overflow (value greater than full scale)
- ☐ bit 6: pulse channel #2 overflow (frequency greater than maximum possible for the given parameters)
- ☐ bit 5: pulse channel #1 overflow (frequency greater than maximum possible for the given parameters)
- ☐ bit 4: measure signal amplitude out of A/D converter range
- ☐ bit 3: measure signal amplitude out of amplifier capability
- ☐ bit 2: input signal error (out of input chain capability)
- ☐ bit 1: coils excitation error
- ☐ bit 0 (LSB) : pipe empty

Meaning and value of the sensor test flags returned with the register 0032:

- ☐ bit 15 (MSB): resistance at electrode E2 is outside the limits respect to the reference value
- ☐ bit 14 : resistance at electrode E1 is outside the limits respect to the reference value
- ☐ bit 13: coil time B is outside the limits respect to the reference value
- ☐ bit 12: coil time A is outside the limits respect to the reference value
- ☐ bit 11: coil temperature is outside the limits respect to the reference value
- ☐ bit 10: coil leakage current is outside the limit
- ☐ bit 09: coil driver output 2 voltage is out of tolerance during test phase 3
- ☐ bit 08: coil driver output 1 voltage is out of tolerance during test phase 3
- ☐ bit 07: coil driver output 2 voltage is out of tolerance during test phase 2
- ☐ bit 06: coil driver output 1 voltage is out of tolerance during test phase 2
- ☐ bit 05: coil driver output 2 voltage is out of tolerance during test phase 1
- ☐ bit 04: coil driver output 1 voltage is out of tolerance during test phase 1
- ☐ bit 03: coil driver power generator voltage is out of tolerance during test phase 2
- ☐ bit 02: coil driver power generator voltage is out of tolerance during test phase 1
- ☐ bit 01: coil driver power generator value is out of tolerance during test phase 2
- ☐ bit 00 (LSB): coil driver power generator value is out of tolerance during test phase 1

Reset totalizers: COMMAND 05.

ADDRESS	SIZE	TYPE OF DATA	COMMAND VALUE	MEANING
0000	1 REGISTER, 16 BITS	UNSIGNED SHORT	0XFF00 (HEX)	Reset the enabled totalizers (same totalizer enabled for reset from digital input).

Diagnostics: COMMAND 08.

ADDRESS	SIZE	TYPE OF DATA	FUNCTION / VALUE
0000	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return query data
0001	1 REGISTER, 16 BITS	SIGNED SHORT	Restart communication
0004	1 REGISTER, 16 BITS	SIGNED SHORT	Activate listen mode
0010	1 REGISTER, 16 BITS	SIGNED SHORT	Clear counters and Diagnostic Register

ADDRESS	SIZE	TYPE OF DATA	FUNCTION / VALUE
0011	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Bus Message Count
0012	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Bus Communication Error Count
0013	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Bus Exception Error Count with CRC error
0014	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Server Message Count
0015	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Server No Response Count
0016	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Server NAK Count
0017	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Server Busy Count
0018	1 REGISTER, 16 BITS	UNSIGNED SHORT	Return Bus Character Overrun Count

MENU 9 - DISPLAY

(POS. 9.1) *Language for all msn* [Language= ITA/EN] **AL1** [LLANG]

Choice of the language. There are 8 languages available: **GB** = English, **IT** = Italian, **TR** = Turkish, **PL** = Polish, **DE** = German, **FR** = French, **PT** = Portuguese, **ES** = Spanish.

(POS. 9.2) *Display Contrast* [Contrast= x] **AL1** [DCNTR]

Display contrast set. The contrast can change according to the room temperature. The allowed range is from 0 to 9.

(POS. 9.3) *KeyBoard Timeout Time* [Disp. time=s xxx] **AL1** [KBTMT]

This function set display/keyboard inactivity. The set values are from 020 to 255 second.

(POS. 9.4) *Display refresh Frequency* [D.rate=Hz xx] **AL1** [DISRF]

Frequency of the display data update. This parameter effects only the display layout and not the response time of the meter itself. The possible choices are: 1/2/5/10 Hz.

(POS. 9.5) *Display function number* [Disp.fn= x] **AL2** [DISFN]

This function sets the display of the page making it visible when you start the display. For each display page is associated with a number that corresponds to the position.

(POS. 9.6) *Display function lock* [Disp.lock= x] **AL2** [DLOKE]

This function locks the scrolling of the display pages selected by the setting. POS. 9.5 see page 29 function.

(POS. 9.7) *Partial totalizer* [Part. tot= ON] **AL2** [PTOTE]

This function enables the display of partial totalizer in visualization pages

(POS. 9.8) *Negative totalizer* [Neg. tot= ON] **AL2** [NEGTE]

This function enables the display of negative totalizer in visualization pages

(POS. 9.9) *Net totalizer* [Net. tot= ON] **AL2** [NVTTE]

This function enables the display of net totalizer in visualization pages

(POS. 9.10) *Date and Time* [Disp.Date= ON] **AL2** [DATDE]

This function enables the display of date and time in visualization pages

(POS. 9.11) *Quick start menu* [Quick start= ON] **AL2** [QSTME]

This function enables the quick start menu.

MENU 10 - DATA LOGGER

(POS. 10.1) <i>Data logger enable</i>	[D.logger en= /OFF]	AL3	[DLOGE]
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This function enables data logger.

The following functions are activated by [D.logger en= ON]

(POS. 10.2) <i>Unite of Measure</i>	[Meas. units= ON]	AL3	[DLUME]
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Measure unit recording enable

(POS. 10.3) <i>Field separator character</i>	[Field separ.= ;]	AL3	[DLFSC]
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This function will set the separator character between data logger data.

(POS. 10.4) <i>Decimal Separator Char.</i>	[Decim.separ.= .]	AL3	[DLDSC]
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This function will set the separator character between data logger number value.

(POS. 10.5) <i>Sample Interval</i>	[Interv.= xx:xx:xx]	AL3	[DLGSI]
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Sampling interval. This function set the log frequency. [Interv.= Hours : Minutes: Seconds]

(POS. 10.6) <i>Total direct totalizer</i>	[Log T+= ON]	AL3	[DTTPE]
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Enable logging of total direct totalizer.

(POS. 10.7) <i>Partial direct totalizer</i>	[Log P+= ON]	AL3	[DTPPE]
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Enable logging of partial direct totalizer.

(POS. 10.8) <i>Total reverse totalizer</i>	[Log T-= ON]	AL3	[DTTNE]
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Enable logging of total reverse totalizer

(POS. 10.9) <i>Partial reverse totalizer</i>	[Log P-= ON]	AL3	[DTPNE]
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Enable logging of partial reverse totalizer

(POS. 10.10) <i>Total Net totalizer</i>	[Log TN= ON]	AL3	[DLTNE]
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Enable logging of total net totalizer

(POS. 10.11) <i>Partial Net totalizer</i>	[Log PN= ON]	AL3	[DLPNE]
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Enable logging of partial net totalizer

(POS. 10.12) <i>Log Flow rate in measure unit</i>	[Log Q(UM)= ON]	AL3	[DFTUE]
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Enable logging of flow rate in measure unit

(POS. 10.13) <i>Log Flow rate in percentage</i>	[Log Q(%)= ON]	AL3	[DFPCE]
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Enable recording of the flow rate as a percentage of full scale value set.

(POS. 10.14) <i>Alarm events</i>	[Log AL.EV= ON]	AL3	[DALEE]
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Enable logging of alarm events

(POS. 10.15) <i>Sensor test result</i>	[Log STR= ON]	AL6	[DSTRE]
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Enable logging of sensor test results

(POS. 10.16) <i>Board temperatures</i>	[Log BTS= ON]	AL6	[DBTSE]
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Enable logging of board temperature

(POS. 10.17) <i>Internal board voltages</i>	[Log IBV= ON]	AL6	[DIBVE]
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Enable logging of internal board voltage

(POS. 10.18) <i>Electrodes DC voltages</i>	[Log EDC= ON]	AL6	[DEDVE]
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Enable logging of electrodes DC voltage

(POS. 10.19) <i>Electrodes AC voltages</i>	[Log AEC= ON]	AL6	[DEAVE]
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Enable logging of electrodes AC voltage

(POS. 10.20) <i>Electrodes source impedance</i>	[Log EIZ= ON]	AL6	[DESIE]
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Enable logging of electrodes impedance

(POS. 10.21) <i>Sensor coils values</i>	[Log SCV= ON]	AL6	[DSCVE]
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Enable logging of sensor coils value

MENU 10 - DATA LOGGER: ONLY MCP FUNCTIONS

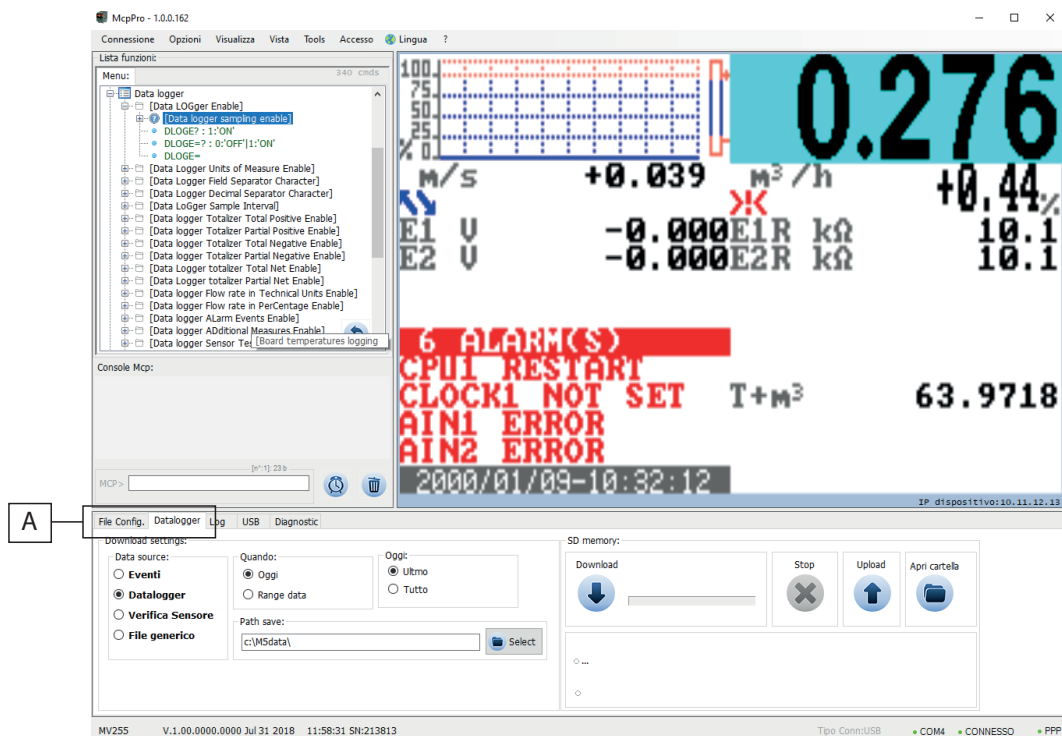
<i>LoG All Information Enable</i>	[MCP ONLY]	AL6	[LGAIE]
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Log all events information. This function save in the event file all MCP commands.

USING DATA LOGGER BY MCP INTERFACE

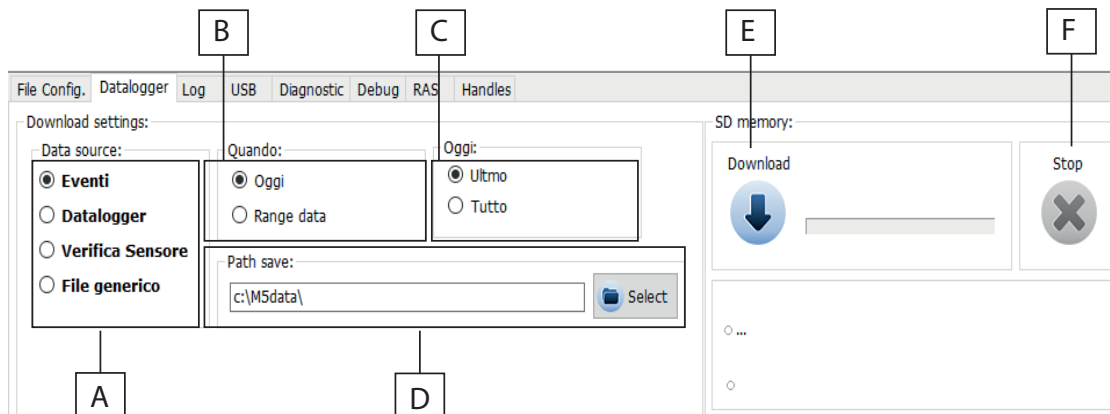
Data are stored on micro SD card; the organization is based on "tree-structure": the system create a daily folder where it save events and data logger . The data can be downloaded by MCP interface.

MCP INTERFACE



Click tab-control data logger to view files.

The sampling data backup depends on the value set by the POS function 10.4 page 25.



A=Data source

Events: Save the file system events (Example F-RAM hardware data [WORKING AREA] [SUCCESSFULLY LOADED])

Data logger: Save files of the enabled data logger function.

Sensor Verify: data logged by BIV function

Generic file:

B=When

Today; It indicates the download file for the current day

Data range; this option allows you to select the date range for download.

C=Today

Last; this option allows to download the latest files, recorder after the last download
All; this option allows the download of all the current day of the file

D=Save path:

This option allows you to save files to the folder on your PC

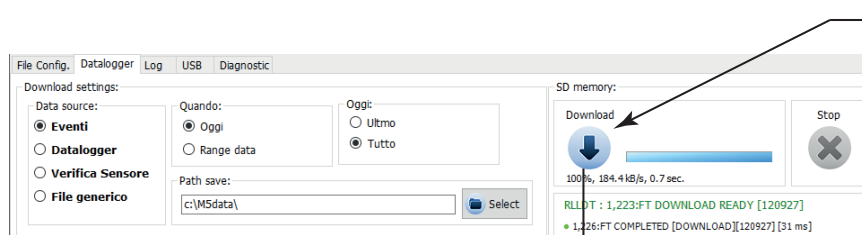
E=Download:

Button to start the download process

F=Stop:

Button to stop the download process

Example: Download Events



Per scaricare tutti gli eventi del giorno odierno in una cartella specifica, settare i seguenti parametri come segue:

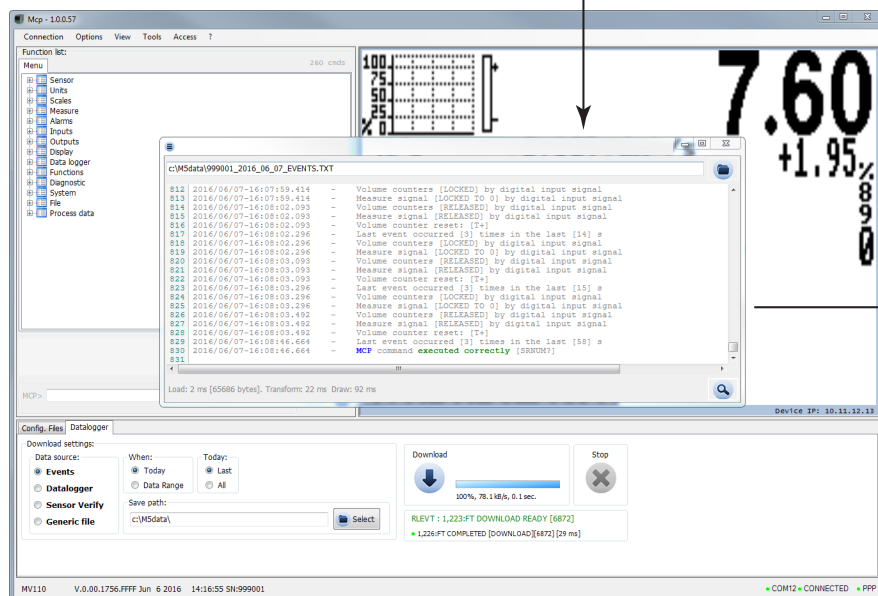
Data source: Eventi

Quando: Oggi

Oggi: Tutti

Save path: C:/.....

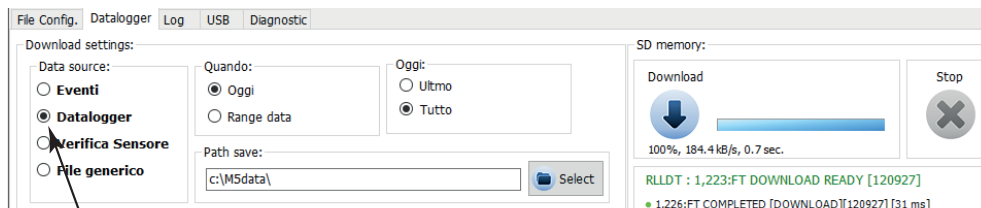
Impostati i parametri cliccare il tasto Download.



Terminato il Download, viene visualizzata una finestra contenente la lista degli eventi in data odierna. Il file viene salvato in formato txt nella cartella specificata per il download.

Example: Download Data Logger

Note: it is recommended the date synchronization between converter and PC to perform correctly the events and logger reading operations.



Per scaricare tutti i dati del giorno odierno in una cartella specifica, settare i seguenti parametri come segue:

Data source: Datalogger

Quando: Oggi

Oggi: Tutti

Save path: C:/.....

Impostati i parametri premere il tasto Download.